Smart Greenhouse Application With MERN and AES-128 CBC Data Encryption Based on IoT

Muhammad Alvito Aditya
Department of Electrical Engineering
UIN Sunan Gunung Djati
Bandung, Indonesia
muhammadalvitoaditya@gmail.com

Nur Rokhman
Department of Electrical Engineering
UIN Sunan Gunung Djati
Bandung, Indonesia
nurrokhman0302@gmail.com

Rin Rin Nurmalasari
Department of Electrical Engineering
UIN Sunan Gunung Djati
Bandung, Indonesia
rinrinnm@uinsgd.ac.id

Sugih Gumilar
Department of Hardware Engineer
PT. Kreasi Rekayasa Indonesia
Bandung, Indonesia
sugihgumilar4@gmail.com

Padlan Alqinsi
Department of Hardware Engineer
PT. Kreasi Rekayasa Indonesia
Bandung, Indonesia
padlanalqinsi@live.com

Nanang Ismail
Department of Electrical Engineering
UIN Sunan Gunung Djati
Bandung, Indonesia
nanang.is@uinsgd.ac.id

2023 9th International Conference on Wireless and Telematics (ICWT) | 979-8-3503-0502-9/23/$31.00 ©2023 IEEE | DOI: 10.1109/ICWT58823.2023.10335284

Abstract* — The greenhouse is a building construction used for growing crops. Many farmers still use traditional methods for the greenhouse data monitoring process, making it inefficient in terms of time and energy spent. Farmers need an attractive, fast, and secure application for an efficient greenhouse data monitoring process. Farmers can use a website-based application connected to the greenhouse's IoT system. This research discusses a website-based smart greenhouse application for the greenhouse data monitoring process. The application was developed using the Scrum Agile Development method, where each stage has an iterative and structured process. The technology combines MongoDB, Express JS, React, and Node JS, known as the MERN framework, the MQTT protocol, and data encryption techniques with AES-128 CBC. Testing this application uses the black box testing method by testing 14 types of REST API services and 9 application services. The duration of testing results varies depending on the number of services to be tested. In addition to being tested with a black box, the application was tested with GTMetrix, which gave an "A" grade with a score of 99%, and the total page size was 519kb. This test proves that MERN has good enough application performance to be implemented as an IoT-based smart greenhouse application. The encryption results of sensor data stored in the MongoDB database have a level of randomness that is very difficult to break, so the data cannot be processed or used to create machine learning models if the developer's key and Initialization Vector (IV) are not known.

***Keywords* — Green House, MERN, MQTT Protocol, AES-128 CBC, Black Box Testing, GTMetrix**

I. Introduction

The greenhouse is a building made of glass or thick plastic material applied to a plant, with translucent construction to manipulate and avoid environmental conditions that are not desired by plants so that plants can produce optimally regardless of the season [1]. However, many farmers still use traditional methods in the greenhouse system for monitoring greenhouse data that affect plants, making it inefficient in terms of time and energy spent. In addition, many farmers still do not pay attention to data security, the speed of the data transmission process, and a fast and attractive application platform for monitoring plant conditions. Therefore, to solve this problem, farmers can use a web-based application with an Internet of Things (IoT) system in the greenhouse, using MERN framework technology (MongoDB, Express JS, React, Node JS), MQTT protocol, and data encryption techniques with AES-128 CBC.

MERN is a collection of frameworks with the same primary programming language, javascript, consisting of MongoDB, Express JS, React/Next JS, and Node JS frameworks. Each framework has different functions, including MongoDB acting as a database, Express JS acting as a backend component, React JS/Next JS acting as a frontend component related to the user interface, and Node JS acting as a runtime environment to run javascript coding on the backend [2], [3]. The research discusses the MERN framework applied to IoT systems, namely the RESTFul IoT service system as a fast communication between IoT devices and the cloud, using the MERN Stack by Preston Porter et al. in 2019 [3].

In addition, the application must also have a data security system by utilizing data encryption technology and an authentication system. Sensor data generated by greenhouse plants is essential because if the data amount to thousands or even millions, machine learning models can be made, such as forecasting weather, plant conditions, and crop yields, which can benefit several parties or large industries. Data stored in the database can be encrypted using the AES-128 CBC cryptographic technique.

The operating mode of the AES-128 CBC cryptographic technique uses a unique initialization vector or array called an IV, the size of the plaintext block (n bits or one plaintext block) to initialize the first block XOR operation [4]. Previously, the plaintext will be divided into several blocks, and later each block can be encrypted with a note that the first plaintext has been encrypted. By using this encryption technique, it is expected to be able to protect large amounts of sensor data in the database in the form of JSON so that others do not easily read it. The research discusses data encryption using AES-128 CBC on IoT systems, with the longest encryption time of 398 microseconds and the smallest throughput of 301507.538 bits/second [5].

In previous research, in 2020, Rafi Dhega Ramanda et al. conducted research with a tool designed for an Internet of Things-based smart greenhouse interface application on Android mobile devices [6]. In 2020, Agus Sofwan et al. conducted research with a tool designed for a greenhouse plant system using an Internet of Things system based on the MQTT protocol [7]. In 2019, Dania Eridani et al. researched the MQTT protocol's performance by implementing a monitoring and controlling system on a Green House Chili plant prototype [8].

979-8-3503-0502-9/23/$31.00 ©2023 IEEE

From the above review, we designed a smart greenhouse system with MERN and AES-128 CBC data encryption based on the IoT. The application development process uses the agile scrum development method, where each software development process can be done iteratively and structured in each part, such as application design, code development, and quality assurance. This system consists of MERN (MongoDB, Express JS, React / Next JS, Node JS), which acts as a framework for creating IoT websites (frontend, backend, and database), MQTT publish-subscribe protocol as a data communication path, and AES-128 CBC data encryption for sensor data security. The microcontroller used in this research is Raspberry Pi 3B, with several components such as a soil moisture sensor, Soil PH, DHT11, oxygen sensor, TDS, water level, lamp, valve, blend, and pump. Thus, this system is expected to produce a secure, attractive, fast IoT website for monitoring and controlling plant data.

II. System Design

A. System Design of Smart Green House Application

The system design in this research is shown in Fig 1. Based on MERN frameworks, the smart greenhouse system consists of 2 main parts: the front and backend. Frontend is the interface application part that users can access using a browser. The backend is the system part that deals with the data behind the screen. Some other parts that work on the backend side are the database and system hardware. The backend has a critical role in performing the data transfer process, authentication system, providing MQTT topics, real-time data connection features with socket io, connection with MongoDB database, and API. All sensor data is stored in the MongoDB database and can be displayed on the frontend side by performing a data description process based on predefined keys and initialization vectors (IV). This process will pass through the REST API as a link between the database and the front end, provided on the backend side containing the syntax for the data transfer process. Data from the sensor is encrypted using the AES-128 CBC data encryption scheme and sent to the sensor using the MQTT publish-subscribe protocol. The frontend and backend communicate through a runtime environment that serves as a code execution process on the frontend and backend using javascript code.

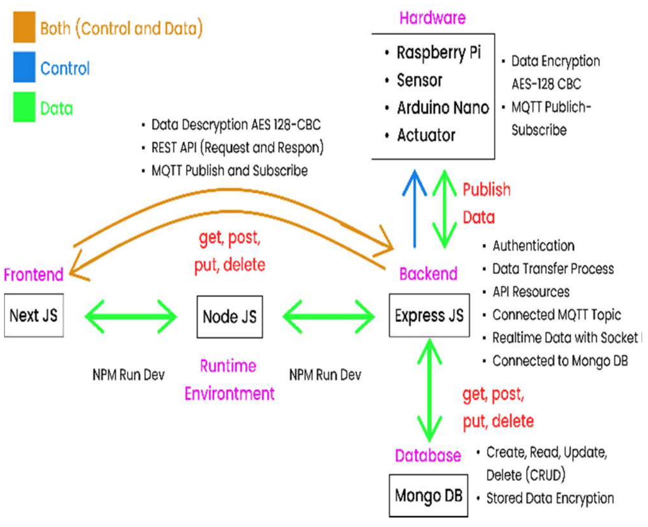

Fig. 1. Design of smart green house application.

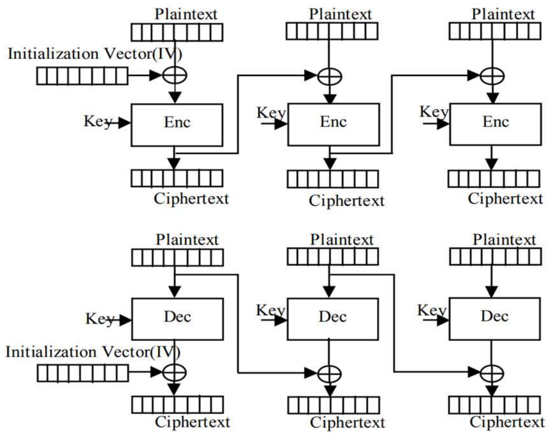

Fig. 2. Cipher Block Chaining (CBC) AES-128 encryption.

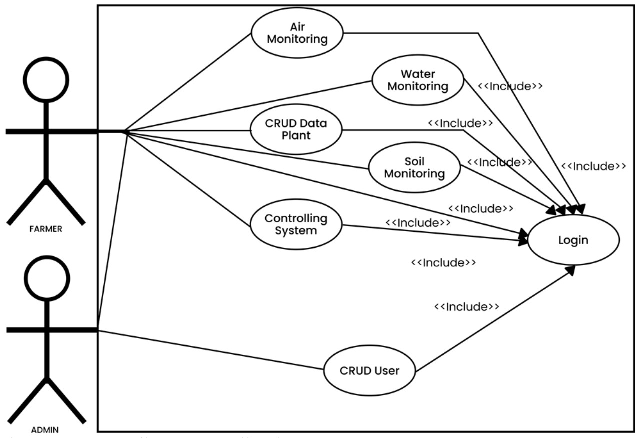

Fig. 3. Usecase diagram application.

B. Data Encryption with AES-128 CBC in IoT Systems

The encryption technique in this research uses the AES-128 method with the Chiper Block Chaining (CBC) operation mode. The AES-128 encryption and description process algorithm consists of 4 main transformation functions in each round: RoundKey, SubBytes, ShiftRows, and MixColoumn [9]. Fig 2 [4] shows the algorithm of the CBC technique in AES-128, where each block generated from the plaintext will be XORed with the encrypted data result, the previous block's ciphertext. The advantage of this technique is that this mode of operation uses an initialization vector IV that is the size of the plaintext block (n bits or one plaintext block) to initialize the first block XOR operation [3]. This makes every cipher message encrypted with CBC unique.

C. Usecase Diagram Application

The use-case diagram for this application is shown in Fig 3. This diagram describes the functionality of the system and the actors who can access the system. This system has login functions, soil monitoring, water monitoring, air monitoring, controlling, Create, Read, Update, and Delete (CRUD) plant data, and CRUD users. There are 2 actors in this application, namely admin and farmers, who have their respective roles in using it. Actors who want to access information through the API and application dashboard must pass the login system with the middleware process so that the jwt token for API access will be obtained and the actor's role will be known. Farmers can access all plant information, sensor data, and control system components such as lights and pumps. Meanwhile, besides being able to access the functions above, the admin actor also plays a role in carrying out the user CRUD process. If the user does not have the admin or farmer status role, the page after the login page will directly redirect to the welcoming page, and the data on the API cannot be displayed as JSON data.

III. Results and Analysis

The smart greenhouse application that has been made is shown in Fig 4, which contains a login page, a dashboard menu containing a recap feature of various sensor data, and a component control page for the smart greenhouse system.

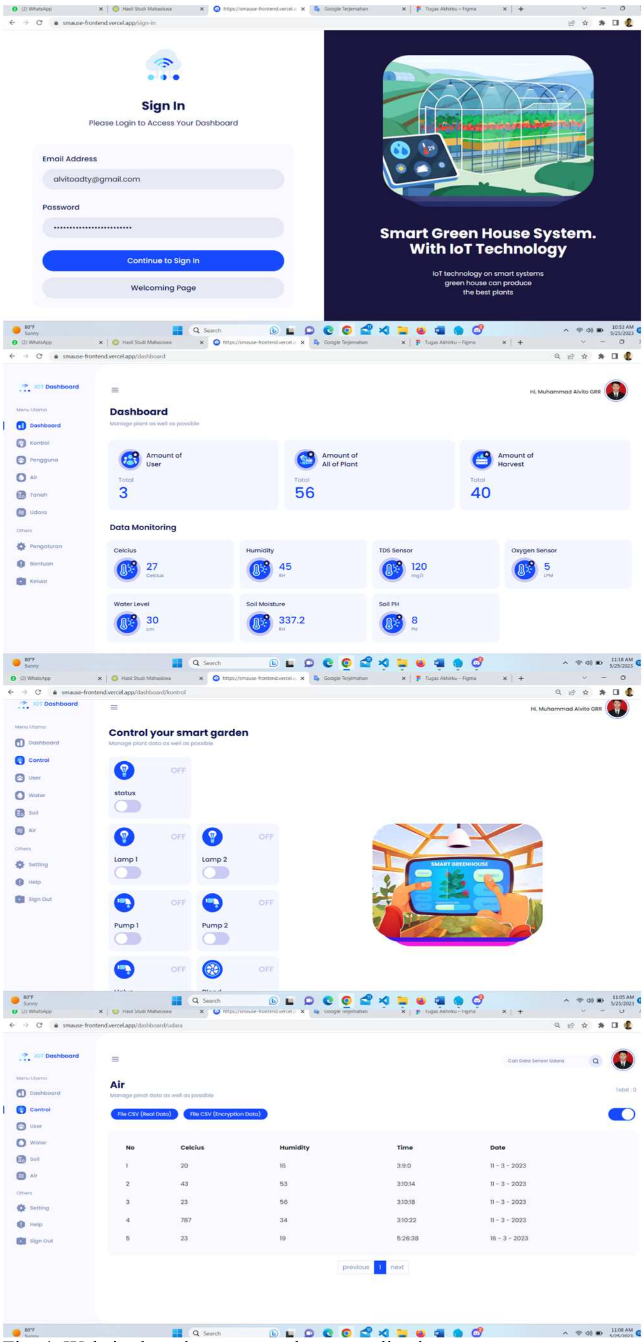

Fig. 4. Website-based smart greenhouse application.

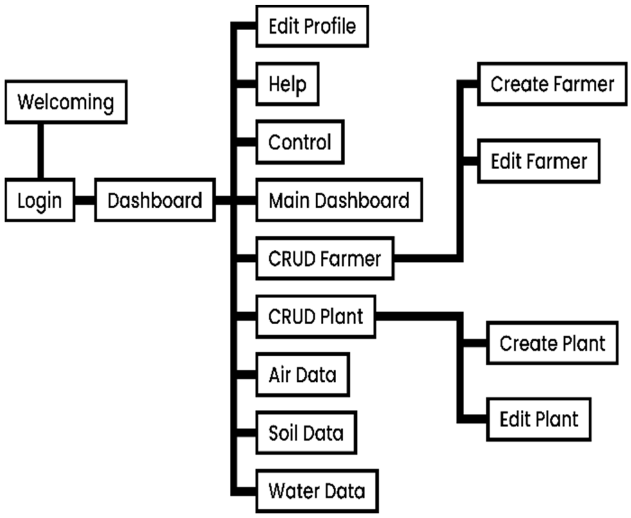

Fig. 5. Application menu structure.

The page menu structure in this application is shown in Fig 5, which contains all pages with various services. The number of web pages in this application is 15 pages. To access the dashboard page, a user must enter the login page first to get a token and access the dashboard middleware. On the CRUD farmer and plant pages, there are 2 additional pages, namely the create page and the edit data page, to add and edit plant data or user data.

Every page view in the application can be accessed publicly because it is hosted with VPS and Vercel. Hosting on VPS adds NGINX and SSL-certified features to add security to the backend system (HTTPS). In addition, to limit access to applications and APIs, there is a middleware system in the form of JWT tokens and admin or farmer roles to maintain and secure application access.

Two aspects are reviewed in this study, namely the MERN-based website aspect, which is tested using the black box testing method, and the data security aspect by comparing the original data from the sensor with the encrypted data using the AES-128 CBC method.

A. Black Box Unit Testing Using Katalon and GTMetrix

The application testing process uses Katalon and GTMetrix tools. Katalon tools are used to test application system functionality, while GTMetrix in Fig 6 is used to measure website performance based on existing features and components.

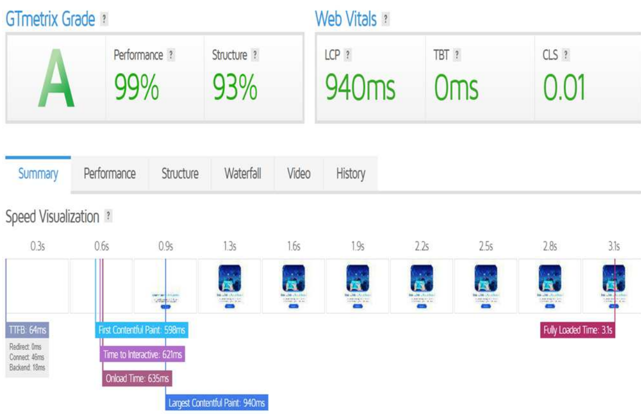

Fig. 6. Performance of application with GTMetrix.

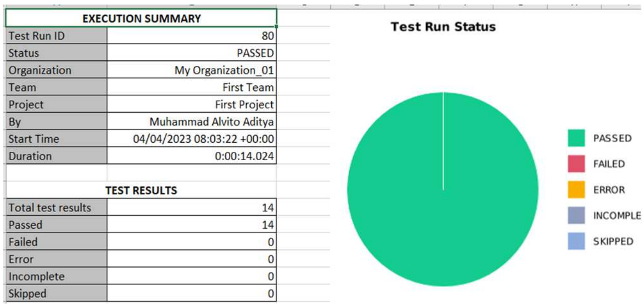

Fig. 7. REST API test results with Katalon.

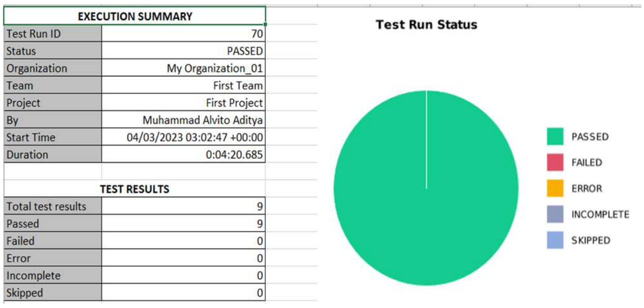

Fig. 8. Application service test results with Katalon.

The results of the summary execution and test run status of the application using Katalon are shown in Fig. 7 and Fig. 8. From this data, it can be seen that the results of application testing that was done previously using the black box testing method, based on the stored repository object records. This tool can store input data, buttons, navigation, etc. so that when running, the website can run automatically based on the repository. If it matches the repository object and status code responses, the application testing process runs smoothly (successfully).

This application testing process uses the black box testing method by testing 14 types of REST API services and 9 application services. The test results show a passed or successful status, as shown in Fig 7 and Fig 8. The duration of the testing results varies depending on the number of services to be tested. In addition to being tested with a black box, this application is tested with the GTMetrix tool in Fig 6, which gives an "A" grade with a score of 99%, and the total page size is 519kb. Several things, such as page rendering, image size, and website algorithm code structure, influence application performance. Farmers can directly use this application to monitor and control greenhouse data because the black box testing process is successfully carried out.

B. The Results of Encyption and Description with AES-128 CBC

The results of security testing of data from sensors in this study are shown in Table I. Data from the original sensor is sent using the AES-128 CBC data encryption scheme. Sensor data will be stored in the form of encrypted data in the MongoDB database, which later, this data can be displayed in the form of original data (data has been encrypted) on the smart greenhouse website so that farmers can monitor and control data. All stored data cannot be used for other needs, such as forecasting if it does not have the key and iv of the AES-128 CBC algorithm. By implementing AES-128 CBC data encryption, the data generated by the smart greenhouse system can be more secure from data thieves.

TABLE I. SENSOR TEST RESULTS WITH DATA ENCRYPTION

TDS Sensor		
No	Original Data	Encrypt Data (AES-128 CBC)
1	4.0	c386f8b9621ef9cc7be95e16cbfe860d
2	2.0	43e65f1428781c6288ea0f8216d2e236
3	8.0	3f51a5c14684d9cd0e192531e4ad8cf0
Oxygen Sensor		
No	Original Data	Encrypt Data (AES-128 CBC)
1	0.1	b67ee0f7787785b6b6ae00aa8f1a74d5
2	19.0	24c1899c151010e38f3cb212626fd8c4
3	10.0	f15931093862045b388b9fa948705259
Water Level Sensor (Ultrasonic)		
No	Original Data	Encrypt Data (AES-128 CBC)
1	33.0	5d4eceb84ce3a1d7288a9b3029d1abd0
2	33.0	5d4eceb84ce3a1d7288a9b3029d1abd0
3	34.0	fe3582dace50230e84ae33c39496d5af
Humidity (DHT11)		
No	Original Data	Encrypt Data (AES-128 CBC)
1	28.9	89425c7ad543d122b55fdc0684a05310
2	29.3	948a7fc633e00cefa64d61e1dba1174a
3	29.3	948a7fc633e00cefa64d61e1dba1174a
Celcius (DHT11)		
No	Original Data	Encrypt Data (AES-128 CBC)
1	58.0	b0aeadb147c4af2df85d95564036cf1a
2	58.0	b0aeadb147c4af2df85d95564036cf1a
3	58.0	b0aeadb147c4af2df85d95564036cf1a
PH Soil PH Sensor		
No	Original Data	Encrypt Data (AES-128 CBC)
1	9.86	fae34dd19e8e3c7d80b3f3bf3f07c80d
2	9.85	dd2f5acad550bc9dbddb8eab1b200de2
3	9.84	d168c72551770465bbf3bd28b865084e
Soil Moisture Sensor		
No	Original Data	Encrypt Data (AES-128 CBC)
1	166.0	9f1539f0f25c071bacfea4823780fbe9
2	225.0	ffe85f19ad0e6af48401890383cc0eae
3	224.0	48fc63aa4c059591594b1ec6077f3a89

IV. CONCLUSIONS

This research has succeeded in developing the smart greenhouse application with MERN And AES-128 CBC data encryption based on IoT. The developed system can provide plant information on the dashboard application, and farmers can monitor and control data using convenient, fast, and safe applications. The results of testing the application using the black box testing method against 14 types of REST API services and 9 application services were declared successful. The duration of testing results varies depending on the number of services to be tested. In addition to being tested with a black box, the application was tested with GTMetrix, which gave an "A" grade with a score of 99%, and the total page size was 519kb. This test proves that MERN has good enough application performance to be implemented as an IoT-based smart greenhouse application. Sensor data stored in the database has been appropriately encrypted using the AES-128 CBC method. Furthermore, we would like to convey that this research will be further developed so that research data analysis will be developed.

ACKNOWLEDGMENT

On this opportunity, we thank the Chairperson of LP2M and the Dean of the FST of UIN SGD Bandung for supporting this research and publication.

References

[1] M. Sc. Ir. Siti Bibah Indrajati, *Persyaratan Teknis Pembangunan Green House Sarana Budidaya Florikultur*. Jakarta: Direktorat Buah dan Florikultura Direktorat Jenderal Hortikultura Kementerian Pertanian Tahun 2022, 2022.

[2] V. Subramanian, *Pro MERN Stack*. Apress, 2019. doi 10.1007/978-1-4842-4391-6.

[3] P. Porter, S. Yang, and X. Xi, "The Design and Implementation of a RESTful IoT Service Using the MERN Stack," in *Proceedings - 2019 IEEE 16th International Conference on Mobile Ad Hoc and Smart Systems Workshops, MASSW 2019*, Institute of Electrical and Electronics Engineers Inc., Nov. 2019, pp. 140–145. doi 10.1109/MASSW.2019.00035.

[4] M. Vaidehi, B. Justus, and R. Principal, "Design and Analysis of AES-CBC Mode for High-Security Applications," in *International Conference on Current Trends in Engineering and Technology, ICCTET'14*, International Conference on Current Trends in Engineering and Technology, ICCTET'14, 2014.

[5] R. Kurniawan Endrayanto, A. Muttaqin, and R. A. Setyawan, "Advanced Encryption Standard (AES) pada Modul Internet of Things (IoT) Advanced Encryption Standard (AES) on Internet of Things (IoT) Module," *TELKA*, vol. 5, no. 2, pp. 103–113, 2019.

[6] R. Dhega Ramanda, A. Sofwan, and D. M. Arfan, "Perancangan Aplikasi Antarmuka Smart Greenhouse Berbasis Internet Of Things (Iot) Pada Perangkat Bergerak Android," *TRANSIENT*, vol. 9, no. 1, pp. 2685–0206, 2020, [Online]. Available: https://ejournal3.undip.ac.id/index.php/transient

[7] A. Sofwan, S. Sumardi, A. I. Ahmada, I. Ibrahim, K. Budiraharjo, and K. Karno, "Smart Greetthings: Smart Greenhouse Based on Internet of Things for Environmental Engineering," in *Proceeding - ICoSTA 2020: 2020 International Conference on Smart Technology and Applications: Empowering Industrial IoT by Implementing Green Technology for Sustainable Development*, Institute of Electrical and Electronics Engineers Inc., Feb. 2020. doi: 10.1109/ICoSTA48221.2020.1570614124.

[8] D. Eridani, "MQTT Performance as a Message Protocol in an IoT-based Chili Crops Greenhouse Prototyping," in *2019 4th International Conference on Information Technology, Information Systems and Electrical Engineering (ICITISEE)*, Yogyakarta, 2019.

[9] S. Arrag, "Design and Implementation A different Architectures of mix column in FPGA," *International Journal of VLSI Design & Communication Systems*, vol. 3, no. 4, pp. 11–22, Aug. 2012, doi: 10.5121/vlsic.2012.3402.